

General Coarse Ghostbusting from a One-Entourage Metric Core

Solution-status note for arXiv:2606.11586

22 July 2026

Status. Literature-implied answer, full scope; likely valid and requiring human review. The existence question has an affirmative answer with $p = 2$. The decisive external input is Roe–Willett’s metric theorem; the reduction to its hypotheses is given in full below.

Source question

Chung and Du ask on page 56 of *Ideal structure of ℓ^p uniform Roe algebras* whether a uniformly locally finite coarse space $(X, \mathcal{E})$ without property A must admit a non-trivial ghost in $B_u^p(X)$ for some $p \in \{0\} \cup [1, \infty]$ [1]. Section 6 has already imposed unitality, and their definition of a coarse structure includes every finite subset of $X \times X$.

ghosts in $B_u^p(X)$ are trivial ghosts, i.e., the ideal I_G^p of all ghosts coincides with the smallest geometric ideal $B_u^p(X, \mathcal{E}_{\min}) = K(\ell^p(X)) \cap B_u^p(X)$.

Proof. It is obvious from Remark 6.5(iv). □

Remark 6.28. Combining the two results yields that property A of the coarse space implies all ghosts in the ℓ^p uniform Roe algebra are trivial. This is consistent with [7] Proposition 2.4] when $p \in [1, \infty)$ and (X, d) is a discrete metric space with bounded geometry and equipped with the bounded coarse structure $(X, \mathcal{E}_d)$.

A natural question is whether for a uniformly locally finite coarse space $(X, \mathcal{E})$ without property A, there exist non-trivial ghosts in $B_u^p(X)$ for some $p \in \{0\} \cup [1, \infty]$? If it is true, then the following are equivalent:

- (i) $(X, \mathcal{E})$ has property A;
- (ii) For some $p \in \{0\} \cup [1, \infty]$, $B_u^p(X)$ has a multiplier approximate identity with controlled propagation;
- (iii) For some $p \in \{0\} \cup [1, \infty]$, all ideals I of $B_u^p(X)$ are geometric ideals;
- (iv) For some $p \in \{0\} \cup [1, \infty]$, all ghosts in $B_u^p(X)$ are trivial ghosts (i.e., $I_G^p = B_u^p(X, \mathcal{E}_{\min})$).

By Theorem 6.23, Theorem 6.25, and Proposition 6.27, we have (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv).

It is known that (iv) $\Rightarrow$ (i) holds for a special case: when $p = 2$, and (X, d) is a discrete metric space with bounded geometry, equipped with the bounded coarse structure $(X, \mathcal{E}_d)$ [32]. In fact, this is not true for $p \in \{0, 1, \infty\}$, which we will show in the next section. However, it remains open for $p \in (1, \infty) \setminus \{2\}$, and a general uniformly locally finite coarse space X .

The source question and its stated general-coarse open scope (arXiv:2606.11586v1, p. 56).

Identification and proof idea

Property A is quantified one entourage at a time. If it fails, one entourage E_0 already witnesses the failure. Retain only the coarse structure generated by E_0 , the diagonal, and the mandatory finite entourages. This smaller structure is countably generated and still fails property A. A direct path-metric construction turns it into a uniformly discrete bounded-geometry metric space without changing its coarse structure. Roe–Willett then produce a non-compact Hilbert-space ghost. The smaller uniform Roe algebra embeds in the original one, while ghostness and non-compactness do not depend on which of the two coarse structures is being viewed.

A metric core for one entourage

Lemma 1 (single-entourage metric core). *Let $(X, \mathcal{E})$ be a countable unital uniformly locally finite coarse space. If $(X, \mathcal{E})$ does not have property A, there is a sub-coarse structure $\mathcal{J} \subseteq \mathcal{E}$ such that:*

- (i) $(X, \mathcal{J})$ is unital, connected, countably generated, and uniformly locally finite;
- (ii) $(X, \mathcal{J})$ does not have property A;
- (iii) there is a uniformly discrete bounded-geometry metric d on X whose bounded coarse structure is exactly $\mathcal{J}$.

Proof. Negating the definition of property A gives an entourage $E_0 \in \mathcal{E}$ and a number $\varepsilon_0 > 0$ for which no controlled support entourage in $\mathcal{E}$ provides the required property-A witness. Let $\mathcal{J}$ be the coarse structure generated by E_0 and Δ_X . It is a substructure of $\mathcal{E}$, is unital, and is connected because, under the conventions of [1], every finite subset of $X \times X$ belongs to every coarse structure. It inherits uniform local finiteness from $\mathcal{E}$. It still fails property A at (E_0, ε_0) : a witness supported in an entourage of $\mathcal{J}$ would also be a witness supported in an entourage of $\mathcal{E}$.

Because $X \times X$ is countable, the mandatory finite entourages form a countable family. Taking all finite coarse operations on that family, E_0 , and Δ_X gives a countable cofinal family in $\mathcal{J}$. Choose an increasing symmetric cofinal sequence $(F_n)_{n \geq 1}$ in $\mathcal{J}$, containing Δ_X , whose union is $X \times X$. For $x \neq y$ put

$$w(x, y) = \min\{n : (x, y) \in F_n\},$$

and set $w(x, x) = 0$. Define $d(x, y)$ as the infimum of the sums of the edge weights along all finite paths from x to y . The weights off the diagonal are positive integers, so the infimum is attained, d is a metric, and $d(x, y) \geq 1$ for $x \neq y$.

For each n , $F_n \subseteq \{d \leq n\}$. Conversely, if $d(x, y) \leq R$ and $N = \lceil R \rceil$, a minimizing path uses at most N positive-weight edges, each lying in F_N ; hence

$$\{(x, y) : d(x, y) \leq R\} \subseteq F_N^{\circ N} \in \mathcal{J}.$$

Cofinality of (F_n) now shows that the bounded coarse structure of d is exactly $\mathcal{J}$. Finally, uniform local finiteness of $\mathcal{J}$ uniformly bounds the cardinality of every d -ball of fixed radius, so d has bounded geometry. $\square$

Full answer

Theorem 1. *Let $(X, \mathcal{E})$ be a countable unital uniformly locally finite coarse space. If $(X, \mathcal{E})$ does not have property A, then $B_u^2(X, \mathcal{E})$ contains a non-trivial ghost. Equivalently, it contains a ghost*

operator that is not compact on $\ell^2(X)$. Thus the source existence question is affirmative, always with $p = 2$.

Proof. Apply Lemma 1 and write $\mathcal{J} = \mathcal{E}_d$. Since property A depends only on the coarse structure, the bounded-geometry metric space (X, d) does not have property A. Roe and Willett’s Theorem 1.3 therefore supplies a non-compact ghost

$$T \in C_u^*(X, d) = B_u^2(X, \mathcal{J})[2].$$

Because $\mathcal{J} \subseteq \mathcal{E}$, every finite- $\mathcal{J}$ -propagation operator has finite $\mathcal{E}$ -propagation. Taking closures in the same operator norm on $\mathcal{B}(\ell^2(X))$ gives

$$B_u^2(X, \mathcal{J}) \subseteq B_u^2(X, \mathcal{E}).$$

The assertion that the matrix coefficients of T vanish at infinity is independent of the coarse structure, and compactness is intrinsic to $\mathcal{B}(\ell^2(X))$. Hence the same T , viewed in $B_u^2(X, \mathcal{E})$, is a non-compact—therefore non-trivial—ghost. $\square$

In [14], Yu introduced *property A*, an amenability property of metric spaces which has since been intensively studied (see [12] for a survey). It is easy to prove that for a property A space of bounded geometry, all ghost operators are compact. Our main objective in this paper is to show

Theorem 1.3. *A bounded geometry metric space without property A always admits non-compact ghosts. That is, property A is equivalent to the property “all ghosts are compact”.*

The decisive metric theorem (Roe–Willett, Theorem 1.3, p. 2).

Scope, provenance, and verification

The result answers the source’s existential question for every space in its Section 6 setting. It does *not* show the corresponding assertion for each fixed $p \in (1, \infty) \setminus \{2\}$; those exponent-specific cases are outside the claim.

This packet is classified as a literature-implied answer rather than as a new full solution. Roe–Willett prove the decisive ghost-existence theorem for metric spaces, while Lemma 1 identifies a metric sub-coarse structure on which their theorem applies. Their paper predates the source question and therefore does not explicitly claim this general-coarse consequence. The source paper itself cites Roe–Willett only for the metric special case.

Checks performed:

- The source question was verified in arXiv:2606.11586v1, p. 56, and Roe–Willett Theorem 1.3 in arXiv:1303.6532v1, p. 2.
- The substructure argument uses the same property-A definition and the same matrix definition of a ghost as the source.

- The metric construction proves equality of coarse structures, avoiding any invariance issue under a merely abstract coarse equivalence.
- No computational or numerical dependency is present.
- On 22 July 2026, the run’s cheap indexes and bounded arXiv searches for the exact source id and combinations of “uniformly locally finite coarse space”, “property A”, “noncompact ghost”, “countably generated”, and “metric” found no explicit publication of this general reduction.

Human-review recommendation. Verify the quantifier negation that produces (E_0, ε_0) and the cofinal-family/path-metric argument in Lemma 1. These are the only steps beyond the cited metric theorem.

References

- [1] Y. C. Chung and X. Du, *Ideal structure of ℓ^p uniform Roe algebras*, arXiv:2606.11586v1 (2026), especially Section 6 and p. 56.
- [2] J. Roe and R. Willett, *Ghostbusting and property A*, J. Funct. Anal. **266** (2014), 1674–1684; arXiv:1303.6532, Theorem 1.3.
- [3] J. Roe, *Lectures on Coarse Geometry*, University Lecture Series 31, American Mathematical Society, 2003. The standard countable-metrization result is Theorem 2.55; an explicit construction sufficient here is included in Lemma 1.